

E-Commerce Data Analysis

A comprehensive SQL-driven analysis of sales performance, customer behavior, product profitability, and time trends across an e-commerce dataset.

\$2.3M

Total Sales

\$286K

Total Profit

793

Total Customers

5K

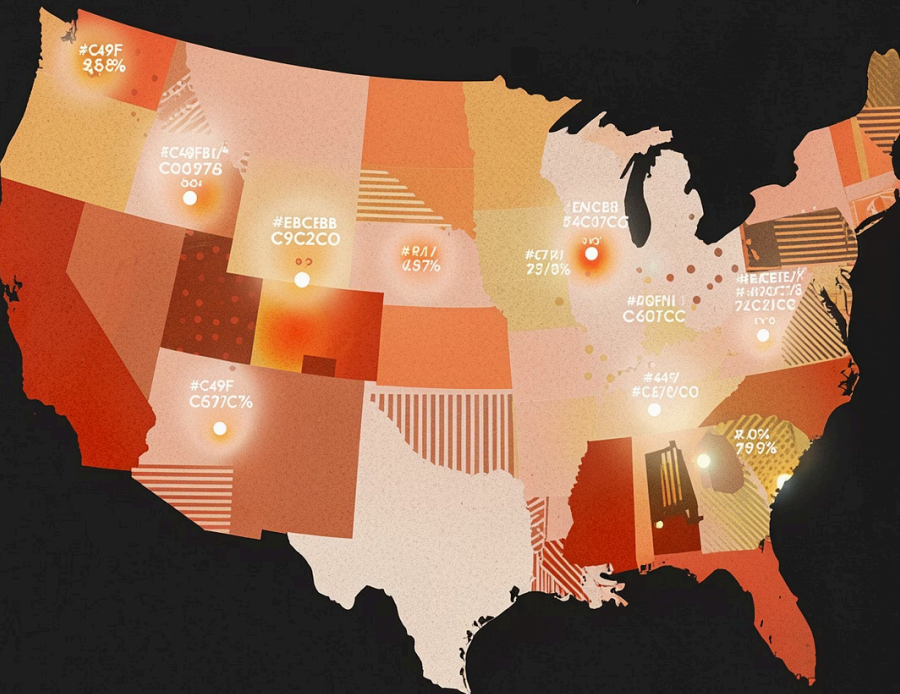
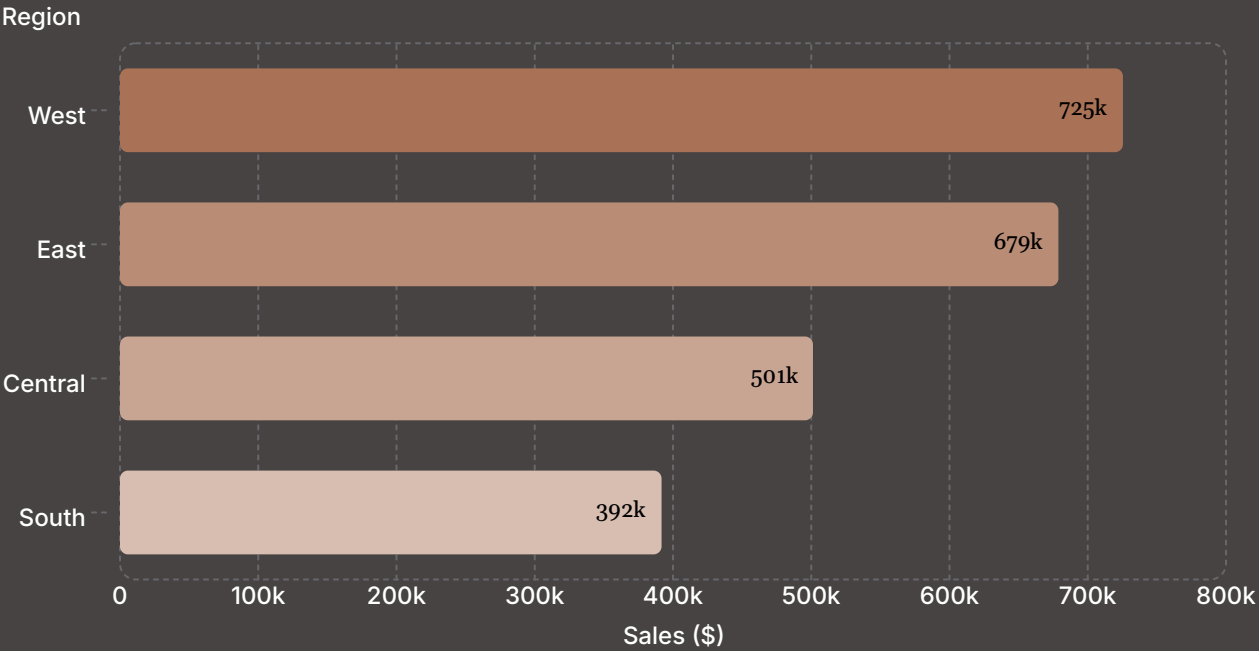
Total Orders

\$230

Avg. Sale

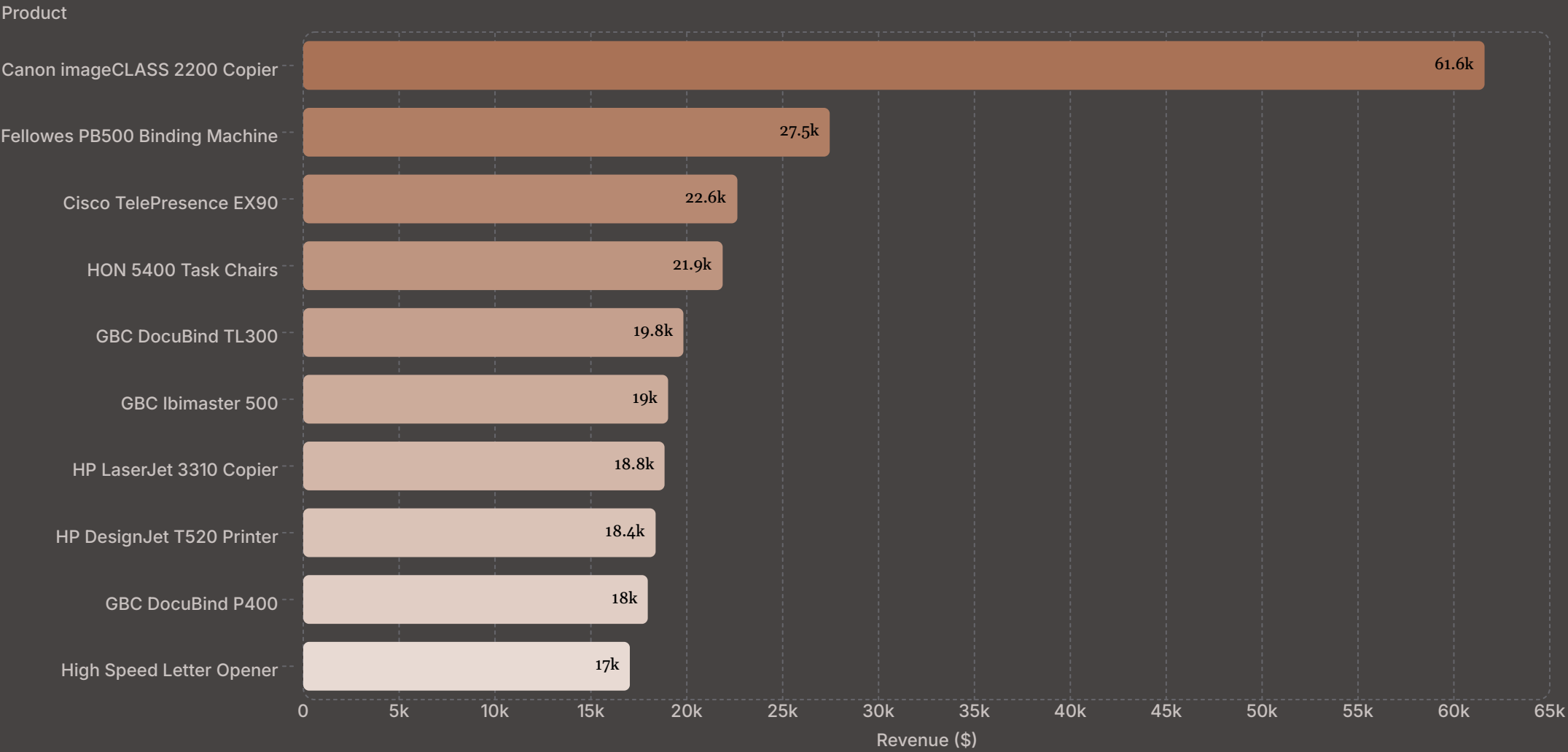
Sales by Region

The **West** and **East** regions dominate, together accounting for nearly **61%** of total sales. Central and South follow at 21.8% and 17.1% respectively.



Top 10 Products by Revenue

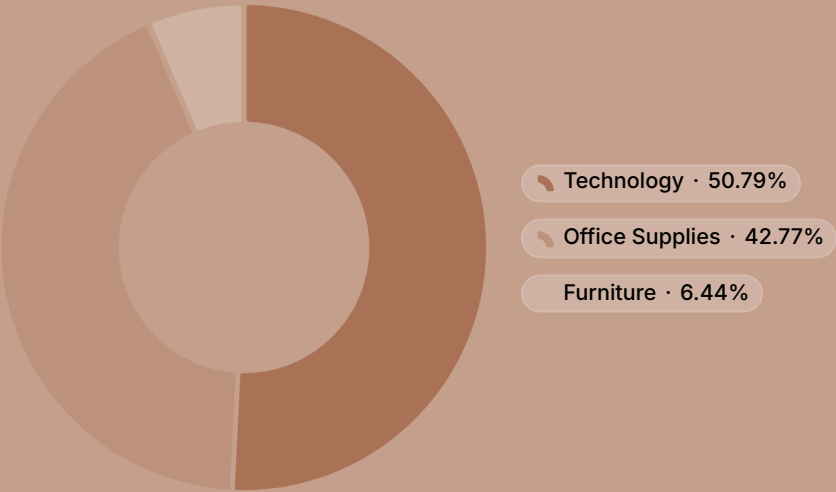
The **Canon imageCLASS 2200 Advanced Copier** leads all products with **\$61,600** in revenue — more than double the second-place product. Binding systems and printers dominate the top 10.



Profitability by Category & Segment

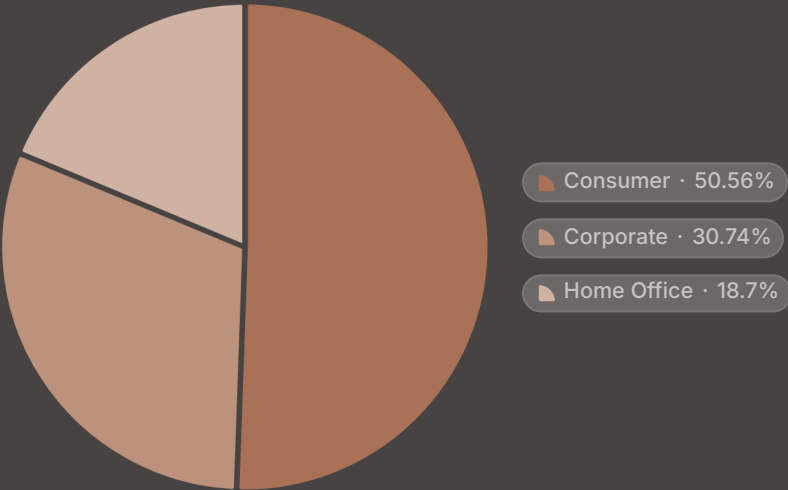
Profit by Category

Technology leads with 50.8% of total profit (\$145K), followed by **Office Supplies** at 42.8% (\$122K). **Furniture** contributes only 6.4% (\$18K).



Revenue by Segment

The **Consumer** segment captures over half of total revenue at 50.6%, followed by **Corporate** at 30.7% and **Home Office** at 18.7%.



Customer Analysis: Spend vs. Profit

High spend does not equal high profit. **Sean Miller**, the top spender at \$25,043, actually generated a **loss of -\$1,981**. The most profitable customer is **Tamara Chand** with \$8,981 profit on \$19,052 in sales — a 47% margin.

Top Spender

Sean Miller — \$25,043 spent, **-\$1,981 profit**

Most Profitable

Tamara Chand — \$8,981 profit, **47% margin**

Best Margin

Jenna Caffey — \$503 profit on \$1,058 spend, **48% margin**

Repeat Customers

88 customers share the top 5 order frequency ranks



Product Profitability: Winners & Losers

✔ Most Profitable Products

Product	Profit (\$)
Canon imageCLASS 2200 Copier	25,200
Fellowes PB500 Binding Machine	7,753
HP LaserJet 3310 Copier	6,984
Canon PC1060 Laser Copier	4,571
HP DesignJet T520 Printer	4,095

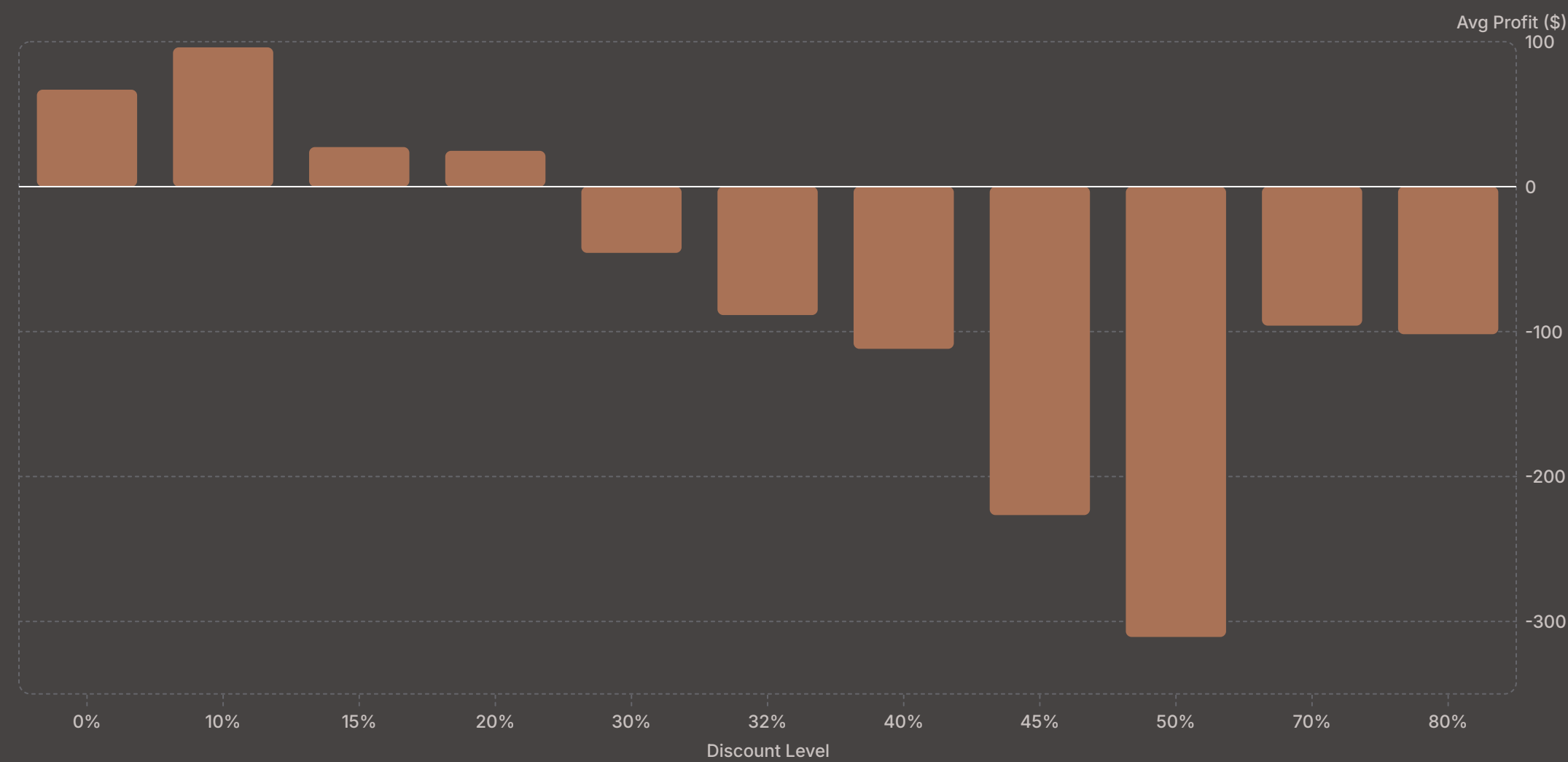
✗ Biggest Loss-Makers

Product	Profit (\$)
Cubify CubeX 3D Printer (Double)	-8,880
Lexmark MX611dhe Laser Printer	-4,590
Cubify CubeX 3D Printer (Triple)	-3,840
Chromcraft Conference Table	-2,876
Bush Racetrack Conference Table	-1,934

① 53 products achieve a **100% profit margin** — meaning the sale price is exactly double the actual cost when no discount is applied.

Discount Impact on Profit

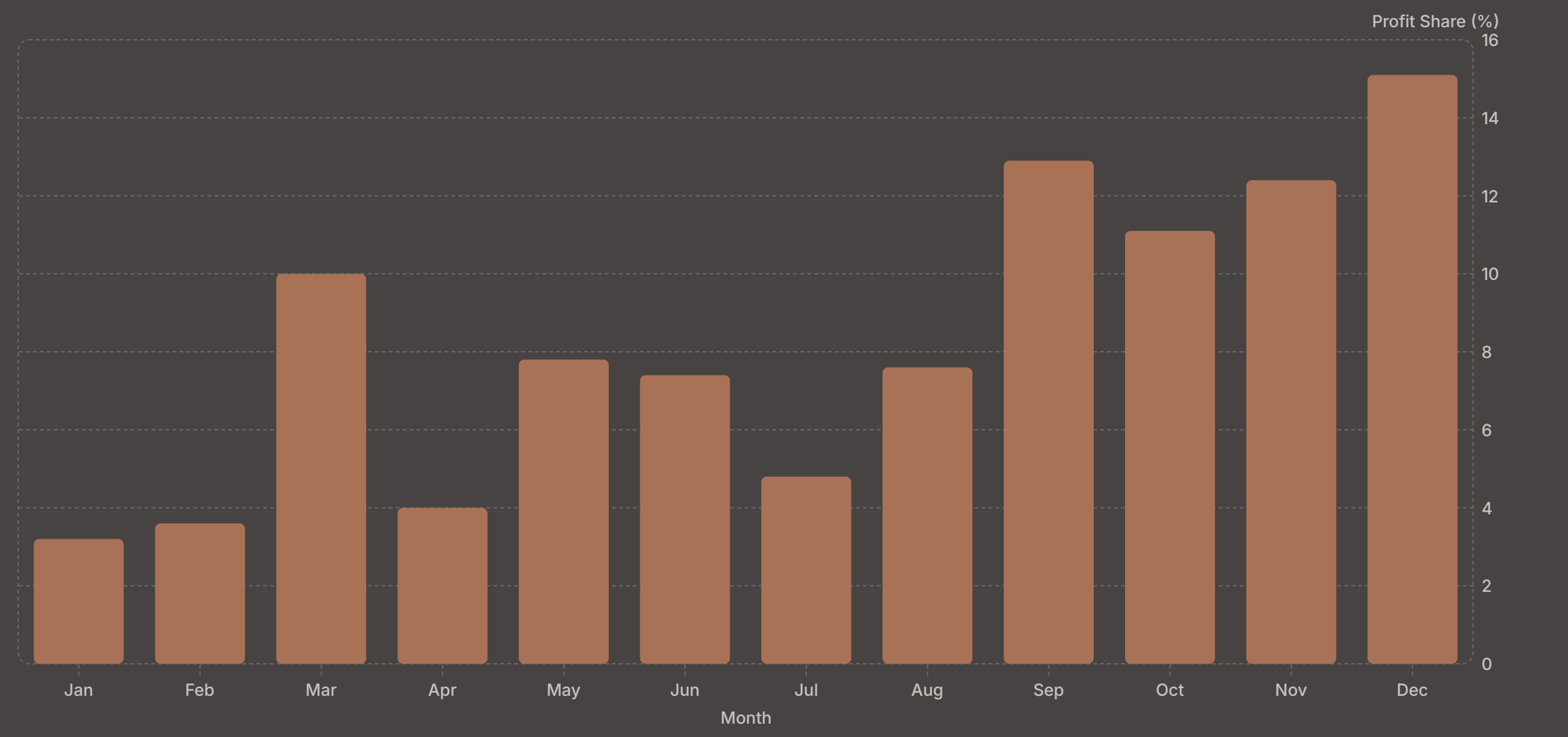
Discounts above **30%** consistently produce negative average profit. At 50% discount, average profit drops to **-\$311**. Binders (37.2% avg. discount) and Machines (30.6%) are the most heavily discounted sub-categories — both exceeding the loss threshold.



⚠ Discounts at or above **30%** result in average losses. Binders and Machines sub-categories are most at risk.

Monthly Sales & Profit Trends

Sales peak dramatically in **Q4**, which accounts for **52%** of annual profit. December alone contributes **15.1%** of total profit, driven by holiday demand. March (10%) and September (12.9%) are secondary peaks.



Shipping Modes & Loss-Making States

Average Shipping Days by Mode

0

Same Day

2

First Class

3

Second Class

5

Standard Class

Same Day and First Class are the fastest options. Standard Class takes up to 5 days on average.

States Generating Losses

State	Loss (\$)
Texas	-25,729
Ohio	-16,971
Pennsylvania	-15,560
Illinois	-12,608
North Carolina	-7,491
Colorado	-6,528

Texas (-\$25.7K) and Ohio (-\$17K) are the largest loss-generating states.



Key Takeaways

- **West & East Drive Revenue**
Together accounting for ~61% of total sales; Texas and Ohio are the biggest loss states.
- **Technology is the Profit Engine**
50.8% of profit; Canon imageCLASS 2200 Copier leads both revenue (\$61.6K) and profit (\$25.2K).
- **High Spend ≠ High Profit**
Top spender Sean Miller generated a loss. Focus on margin, not just volume.
- **Cap Discounts at 30%**
Discounts above 30% consistently produce losses. Binders and Machines are highest risk.
- **Q4 is Critical**
The last quarter generates 52% of annual profit — December alone contributes 15.1%.